

Examiner
only

3420U201
03

Answer **all** questions.

1. (a) The table below contains statements about the motion of an object and the forces acting on it.
For each statement, place **one** tick (✓) to show which law it applies to.
One row has been completed as an example. [3]

Statement	Newton's 1st Law	Newton's 2nd Law	Newton's 3rd Law
A skydiver accelerates when the forces are unbalanced.	✓		
This law can be written as $F = ma$.			
A train will not move from rest unless acted on by a resultant force.			
When it is fired, a rifle exerts a force on a bullet. The bullet exerts an equal and opposite force on the rifle.			

- (b) A ball is dropped from rest ($u = 0 \text{ m/s}$) from a window.
It takes a time, $t = 1.2 \text{ s}$ to reach the ground.

- (i) Use the equation:

$$v = u + at$$

to calculate the speed, v , of the ball as it hits the ground.
(Acceleration, $a = 10 \text{ m/s}^2$) [2]

$v = \dots\dots\dots \text{ m/s}$

- (ii) Use your answer above for v and the equation:

$$x = \frac{u + v}{2} t$$

to calculate the distance, x , of the window above the ground. [2]

$x = \dots\dots\dots \text{ m}$

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